Relativity and Electromagnetism: chapter-by-chapter what to study

	What to study	*Notes*
Chapter 2	Entire chapter!	Capacitance [2.5.4] is not very important for the final exam.
Chapter 3	3.2, 3.3.2, 3.4	Formal aspects in the initial part of the chapter were not covered. Separation of variables was covered only for spherical coordinates.
Chapter 4	4.1.4, 4.2, 4.3, 4.4.1, 4.4.2	Initial part of the chapter [4.1.1, 4.1.2, 4.1.3] is not very important expect for an overall understanding. The subsections which are most important here are bound charges [4.2.1], the ***D*** field [4.3.1], linear dielectrics and permittivity [4.4.1]. Energy in dielectric systems [4.4.3] and forces on dielectrics [4.4.4] can be skipped.
Chapter 5	Entire chapter!	Multipole expansion of the vector potential [5.4.3] can be skipped.
Chapter 6	6.1.4 6.2, 6.3, 6.4.1	Many parts of Chapter 6 are repetitions of Chapter 4. Initial part of the chapter [6.1.1, 6.1.2, 6.1.3] is not very important expect for an overall understanding. The subsections which are most important here are bound currents [6.2.1], the ***H*** field [6.3.1], linear magnetic materials and permeability [6.4.1]. Ferromagnetism [6.4.2] can be skipped.
Chapter 7	7.1.2, 7.1.3, 7.2, 7.3	Inductance [7.2.3] is not very important for the final exam.
Chapter 8	8.1, 8.2.1, 8.2.2, 8.2.3	Maxwell stress tensor [8.2] was covered in class and in homework assignments, but will not be included in final exam.
Chapter 9	9.2.3, 9.3.1	Only some parts of this chapter were discussed in relation to the Poynting vector [9.2.3] and wave propagation in material media [9.3.1].
Chapter 10	10.1, 10.2, 10.3	In 10.2 and 10.3 some calculations were skipped but final results are important.
Chapter 11	11.1.1, 11.1.2	Electric dipole radiation was covered in detail; **Larmor formula** at end of the section was mentioned.
Chapter 12	Entire chapter!	

Relativity and Electromagnetism: important theory sections (some examples)

Note: These are only a few examples, and this is not an exhaustive list of possible theory questions.

1. How $\nabla\times\mathbf{E} = \mathbf{0}$ admits the definition of the **electric potential** (Section 2.3.1).

2. How the Ampère-Maxwell equation (with Maxwell's displacement term) is consistent with the **equation of continuity** (Section 7.3.2, Section 7.3.3, Section 8.1.1).

3. Proof of the **Poynting's theorem** (Section 8.1.2).

4. How the **gauge transformations** leave the fields unchanged (Section 10.1.2).

5. How retarded potentials in the **Lorenz gauge** satisfy the wave equation (Section 10.2.1).

6. For **Liénard-Wiechert potentials** (and the resulting fields), you won't be asked to perform a full end-to-end derivation. However you may be asked to carry out an intermediate step. Interpretation of the final expressions is important. So please remember the quantities (such as *retarded time* and *retarded position*) which appear in the expressions.

7. How **time dilation**, **length contraction**, **Einstein velocity addition, transformation law for forces**, etc. follow from the **Lorentz transformations**.

8. **Covariant electrodynamics**: derivation of familiar version of Maxwell's equations from equations of electrodynamics in tensor notation.

9. Expression for bound charges from **polarization** (Section 4.2.1).

10. For **radiation** (Section 11.1.2), you won't be asked to perform a full end-to-end derivation. However an understanding of the different *approximations* is important and so is the interpretation of the final expressions. You may be asked about some of the intermediate steps.